

RAMBØLL	PROJECT —	PROJECT NO. —	PREPARED CHECKED APPROVED	TE — —
	Concrete Modulus at Age t (EN 1992-1-1 3.1.3) EN 1992-1-1:2004 §3.1.2, §3.1.3			
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PURPOSE

Time-development of the concrete secant modulus of elasticity, from cement class and mean 28-day strength.
Shared upstream of crack-width, deflection and bending-stiffness calculations.

1 Input Parameters

Symbol	Value	Unit	Description
S_{cem}	0.25	-	Cement class coefficient (0.20 R / 0.25 N / 0.38 S)
t	7	days	Concrete age at loading
f_{cm}	38	MPa	Mean compressive strength at 28 days (C30/37)
E_{cm}	32837	MPa	Secant modulus at 28 days (C30/37)

2 Calculation (all formulas from the shared library)

Concrete strength development coefficient at age t

$$\beta_{cc,t} = e^{S_{cem} \cdot (1 - \sqrt{\frac{28}{t}})}$$
$$= e^{0.25 \cdot (1 - \sqrt{\frac{28}{7}})}$$
$$= \mathbf{0.7788}$$

EN 1992-1-1:2004 §3.1.2(6), Eq. (3.2)

Mean concrete compressive strength at age t

$$f_{cm,t} = \beta_{cc,t} \cdot f_{cm}$$
$$= 0.7788 \cdot 38$$
$$= \mathbf{29.59 \text{ MPa}}$$

EN 1992-1-1:2004 §3.1.2(6), Eq. (3.1)

Secant modulus of elasticity at age t

$$E_{cm,t} = E_{cm} \cdot (\frac{f_{cm,t}}{f_{cm}})^{0.3}$$
$$= 32837 \cdot (\frac{29.59}{38})^{0.3}$$
$$= \mathbf{30464 \text{ MPa}}$$

EN 1992-1-1:2004 §3.1.3(3), Eq. (3.5)

3 Verification

OK

Modulus at early age does not exceed the 28-day value
 $E_{cm,t} = 30464 \text{ MPa} \leq E_{cm} = 32837 \text{ MPa}$